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# G H Raisoni International Skill Tech University, Pune

F.Y. B. Tech (TERM-I)

CAE II- WINTER 2025 (2024 Pattern)

## Environmental Chemistry (25UBSL1102)

(Time: - 01 Hour)

(Max. Marks: 20)

29.7
27.3
02

**Instructions to the students:**

*i) All questions are compulsory. ii) Neat diagrams must be drawn wherever necessary.*
*iii) Figures to the right indicate full marks. iv) Use of scientific calculator is allowed*
*v) Assume suitable data, if necessary.*

| Q.No | *Question* | Marks | CO | BL |
|---|---|---|---|---|
| Q1.(a) | Describe the construction & working of Lead-acid battery along with diagram & reactions involved. | [5] | 3 | 2 |
| b) | Differentiate between a primary & secondary battery. | [5] | 3 | 2 |
| | OR | | | |
| Q 2.(a) | Explain the construction & working of Lithium -ion battery along with diagram & reactions involved. | [5] | 3 | 2 |
| (b) | Mention the various components of a battery. Discuss their role & reactions involved. | [5] | 3 | 2 |
| | | | | |
| Q 3.(a) | Describe the construction and working of Bomb calorimeter with neat diagram. Give the expression for the GCV formula using bomb calorimeter. | [5] | 4 | 2 |
| (b) | Calculate the HCV (in kJ/kg) 0.75 gm of a fuel containing 80% of carbon, when burnt in a bomb calorimeter increased the temperature of water from 27.3 to 29.7$^0$ C. The calorimeter contains 250 grams of water and it's water equivalent is 150 grams. 401952 | [5] | 4 | 2 |
| | OR | | | |
| Q. 4 (a) | Explain the construction and working of Biogas plant with suitable diagram. Mention any two application's and advantages. | [5] | 4 | 2 |
| (b) | Observations in the Boy's Calorimeter experiment on gaseous fuel are given below, find GCV and NCV of the fuel.<br>Volume of gas burnt (STP) 0.08m$^3$<br>Mass of cooling water used=29.5 kg<br>Rise in temperature of circulating water=9.1 $^0$ C<br>Mass of condensed steam =0.04kg | [5] | 4 | 2 |